

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign

Jun 2021 - present
Aug 2017 - May 2021

WORK EXPERIENCE

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483)

Jun 2021 - present

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern

May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated builds in TortoiseSVN using Python, reducing need for manual integration.

RESEARCH

Aerodynamics and Planetary Exploration Group (APEX)

Advisor: Dr. Siegfried Eggel

Interstellar Navigation and Star Identification

Sep 2024 - present

- ❖ Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

Celestial Navigation to Supplement DSN Tracking

Jul 2024 - present

- ❖ Collaborating with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- ❖ Designing sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- ❖ Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars

Nov 2022 - present

- ❖ Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- ❖ Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

Aerospace Mission Analysis Laboratory

Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV)

Jun 2021 - present

- ❖ Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- ❖ Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package [PINT](#) to within 50 μ s.
- ❖ Demonstrated the computational feasibility of (XNAV) in a lost in space scenario.

Velocity-based Initial Orbit Determination

Aug 2019 - Apr 2024

- ❖ Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- ❖ Assessed the performance of velocity IOD methods using Monte Carlo simulation.

Space Systems Optimization Laboratory

Advisor: Dr. Koki Ho

Architecture Trade Studies on IRSU Technologies for Human Space Exploration

Sep 2018 - Sep 2019

- ❖ Worked with two graduate students to discuss the development of a mixed-integer linear programming framework for space logistics optimization.
- ❖ Conducted literature review of in-situ resource utilization (ISRU) technologies to quantify input/output rates of technology options for an architecture optimization tool.

SKILLS

Programming Languages	Python, C/C++, MATLAB, Javascript
Software/Prepackaged Tools	L ^A T _E X, Git, GMAT
Operating Systems	Linux, MacOS, Windows
Languages	English, Mandarin Chinese

HONORS AND AWARDS

Yee Fellowship

2021, 2024

Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship

2024

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference

2024

American Astronautical Society (AAS) Rocky Mountain Section